



BIMSARA GUNAWARDANA

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[Bimsara Gunawardana](#) [bimsara0608](#) [Portfolio](#) [Grabcad](#)

SUMMARY

Mechanical and Mechatronics Engineering focused professional with 3+ years of experience in CAD, industrial automation, and embedded systems. Certified SOLIDWORKS Professional (CSWP) with strong skills in design for manufacturing and product development. Delivered 60+ engineering projects and contributed to PLC-controlled machinery at MAS Holdings, improving production efficiency and system reliability. Seeking Mechanical Design, Mechatronics, or Automation Engineering roles.

WORK EXPERIENCE

Autonomation Engineering Intern 2025 (Feb–Aug) | [MAS Holdings – Bodyline](#)

- Engineered and deployed PLC-integrated automation solutions improving process reliability and enabling real-time fault detection in production lines.
- Designed and fabricated mechanical components and performed AGV troubleshooting, supporting downtime reduction in production.

Mechanical Designer (2023 - 2024) | [SEDS Sri Lanka](#)

- Contributed to the Taprobane 4.0 Mars rover by focusing on manipulator arm development, kinematic calculations, and joint design within a multidisciplinary team.

Co-Founder & Lead 3D Designer (2022 - 2024) | [VirtualPensar Pvt Ltd](#)

- Co-founded VirtualPensar Pvt Ltd; led 3D design and prototyping, delivering 60+ projects including functional prototypes and 3D-printable components.

Concept & 3D Designer (2021 - 2022) | [Lautus Robotics](#)

- Developed concept designs and 3D CAD models and participated in manufacturing and assembly of an AGV for the Civil Aviation Authority of Sri Lanka.

EDUCATION

University of Colombo (2022 - 2026)

Bachelor of Engineering Technology Honours in Instrumentation and Automation

Relevant Coursework

Embedded Systems, Robotics,	3D Modeling and CAD, FPGA,	Algorithms, CFD & FEA,	Micro-controllers, PLC Programming
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Sri Lanka Institute of Robotics (SLIR) (2022 - 2023)

PLC Programming (Completed)

Nalanda College Colombo

- G.C.E Advanced Level - Technology stream - 4A passes with a Z - score of 2.6405 (Colombo district 11)
- G.C.E Ordinary Level - 8A passes and 1B pass (including A in English and Mathematics)

PUBLICATIONS

Development of a Parallel Arm Robot for Seed Handling Applications in Agriculture

U.V.H. Sameera, P.K.T. Akmal, W.G.K.S. Buddhika, B.M.K.S.T. Chandra, **G.G.B.S. Gunawardana**, A.V.W. Jayawardana, U.S. Mutugala

Presented at the Annual Research Symposium 2024, University of Colombo, Sri Lanka, p. 352.

PROJECTS

Maritime Search and Rescue (SAR) drone - University Final Year Project

- Leading the development of a drone, integrating computer vision algorithms, CFD-optimized aerodynamics, and a custom-engineered deployable payload system with real-time communication.

Taprobane 4.0 Mars Rover Project - Associated with SEDS Sri Lanka [↗](#)

- Designed and simulated a high-efficiency cycloidal gear elbow joint using Autodesk Inventor to ensure precise, low-backlash robotic manipulation.

Smart Mailbox – Sahasak Nimavum 2023 Silver Award [↗](#)

- Led a team to develop an award-winning smart mailbox system, providing real-time notifications and secure parcel management.

Autoguided Fabric Feeding system Development - Associated with MAS Holdings

- Designed and manufactured a conveyor based fabric feeding system integrating custom-manufactured mechanical components with PLC-controlled automation for synchronized operation.

Real-Time Multithread Monitoring System - Associated with MAS Holdings

- Designed and developed an embedded thread continuity detection system with custom PCB design, real-time control logic, and deployment with user documentation for factory use.

Automated Runway Cleaning Rover - Associated with Lautus Robotics

- Contributed to the initial phase of developing an autonomous guided vehicle aimed at cleaning airport runways, focusing on foundational design and prototyping.

3D Printing and Design Marketplace Platform

- Leading the development of a real-time 3D printing platform that enables automated pricing for models 3D prints and an integrated marketplace for hiring professional designers.

TECHNICAL SKILLS

Certifications : Certified SOLIDWORKS Professional (CSWP), Credential ID C-29RCXDMEHU [↗](#)

3D Design/Simulation: SolidWorks, AutoCAD, Autodesk Inventor Pro, Ansys, Blender

Programming : Python, C, HTML, CSS, JavaScript, PHP, VHDL, MATLAB, Ladder Logic, SQL

Developer Tools: VS Code, EasyEDA, Altera quartus ii, Git, Atmel Studio

Technologies/Frameworks: React, Arduino, ESP32, Raspberry Pi, AWS

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Volunteer

- Chief Coordinator (2025 - 2026) Society of Instrumentation and Automation Technology (SIAT)
- Member of SEDS UoC (University of Colombo Branch of SEDS Sri Lanka)
- Member of the Web development team - Arduino day 2025
- Vice President, Nalanda College Aeronautical Society (2018 – 2019)

Activities

- Member of University Swimming Squad
- Secured 1st runner-up positions in the Freshers' and Inter-Faculty Badminton Championships.

REFERENCES

Dr. B. L. Sanjaya Thilakarathne

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and Automation Technology,
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